

SHINYMR: A SHINY APP FOR RIGOROUS AND REPRODUCIBLE MENDELIAN RANDOMIZATION

New Researchers Conference

Fred Boehm

Thursday, July 31, 2025

COLLABORATIVE RESEARCH WITH JI HOON PARK



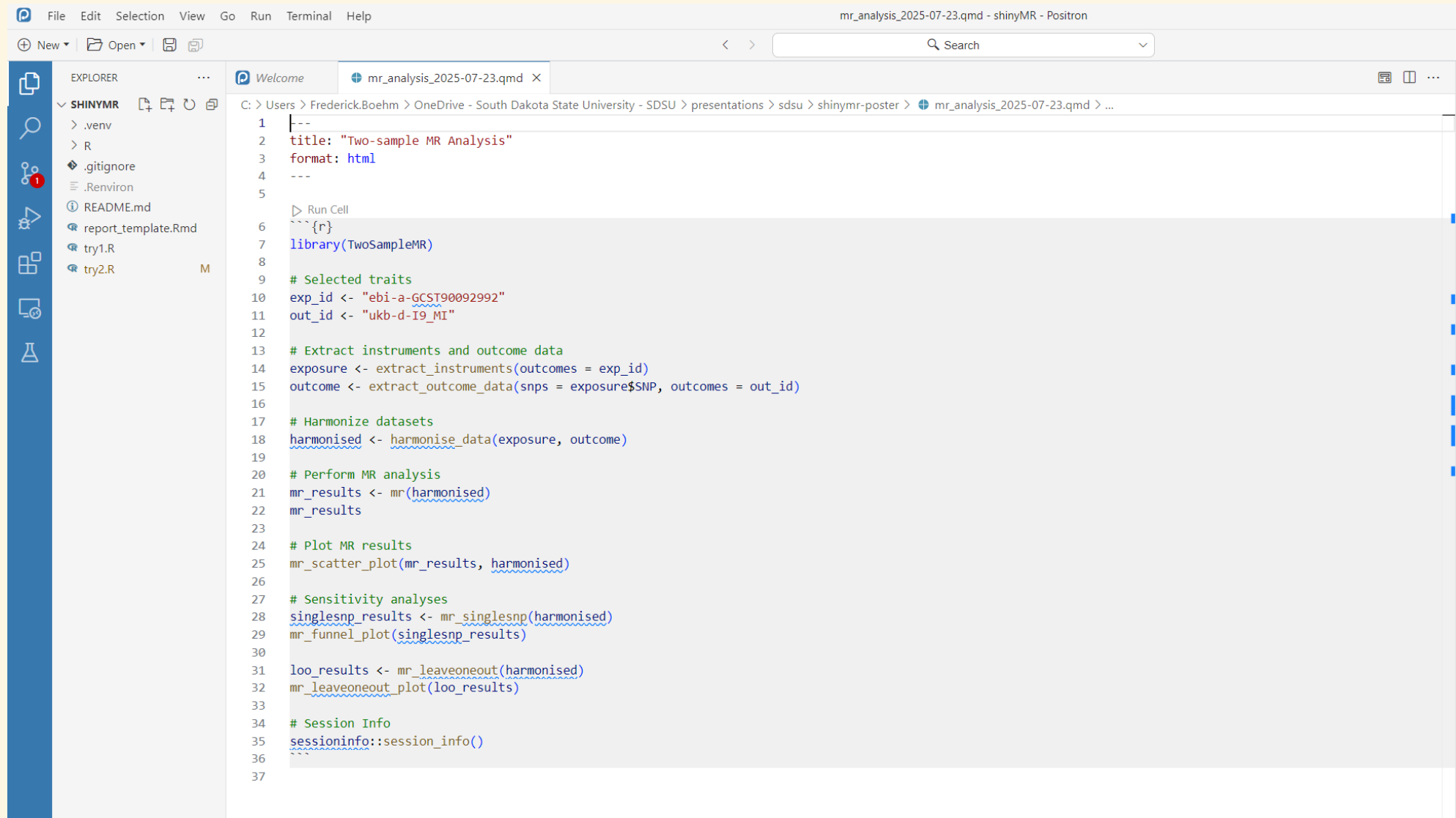
MOTIVATION FOR A SHINY APP

- Many recent reports have questioned the scientific rigor of published MR studies
- To promote rigor and reproducibility, shinyMR users automatically perform MR analysis and sensitivity analyses to evaluate the robustness of MR assumptions

SHINY APP DESIGN CONSIDERATIONS

- Interface with openGWAS database
- Require sensitivity analyses to assess MR assumptions
- Users must manually ensure that there is no subject overlap in the two GWAS

DOWNLOAD A REPRODUCIBLE QUARTO DOCUMENT



The screenshot displays the Positron IDE interface. The top menu bar includes File, Edit, Selection, View, Go, Run, Terminal, and Help. The title bar shows the file name: mr_analysis_2025-07-23.qmd - shinyMR - Positron. The left sidebar features an Explorer panel with a file tree for the project 'SHINYMR', including files like .venv, R, .gitignore, .Renvirom, README.md, report_template.Rmd, try1.R, and try2.R. The main editor area shows the content of the selected file, mr_analysis_2025-07-23.qmd, which is a Quarto document. The document content includes a title, format, and a series of R code blocks for MR analysis, such as library(TwoSampleMR), data extraction, harmonization, MR analysis, and plotting.

```
1  {--
2  title: "Two-sample MR Analysis"
3  format: html
4  ---
5
6  Run Cell
7  {r}
8  library(TwoSampleMR)
9
10 # Selected traits
11 exp_id <- "ebi-a-GCST90092992"
12 out_id <- "ukb-d-I9_MI"
13
14 # Extract instruments and outcome data
15 exposure <- extract_instruments(outcomes = exp_id)
16 outcome <- extract_outcome_data(snps = exposure$SNP, outcomes = out_id)
17
18 # Harmonize datasets
19 harmonised <- harmonise_data(exposure, outcome)
20
21 # Perform MR analysis
22 mr_results <- mr(harmonised)
23
24 # Plot MR results
25 mr_scatter_plot(mr_results, harmonised)
26
27 # Sensitivity analyses
28 singlesnp_results <- mr_singlesnp(harmonised)
29 mr_funnel_plot(singlesnp_results)
30
31 loo_results <- mr_leaveoneout(harmonised)
32 mr_leaveoneout_plot(loo_results)
33
34 # Session Info
35 sessioninfo::session_info()
36
37
```

THANK YOU!